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Configuring SSH Remote Access on Ubuntu Server

Objective

Set up an Ubuntu Server with SSH (Secure Shell) remote access and connect to it from a host desktop machine using Bridged Network mode.

Operating System Installation

Install Ubuntu Server

1. Start the VM and boot from the Ubuntu Server ISO
2. Follow the installation wizard:
 - Profile setup: Create a user account (e.g., server / server123)
 - Install OpenSSH server: **Check this option** (Important!)
 - Selected software: **None** (Standard Ubuntu Server)
3. Wait for installation to complete and reboot

Network Configuration

Configure Network Mode

For VirtualBox:

1. Power off the VM (if running)
2. Go to **Settings** → **Network**
3. **Adapter 1**: Enable Network Adapter
4. **Attached to**: Select **Bridged Adapter**
5. **Name**: Select your host's physical network adapter (Wi-Fi/Ethernet)
6. Click **OK**

For VMware:

1. Power off the VM
2. Go to **VM Settings** → **Network Adapter**

3. Select **Bridged** mode

4. Click **OK**

Install/Verify SSH Server

Server: Install SSH on the server (if not already installed):

- `sudo systemctl status ssh` – for checking the SSH status.
- `sudo apt update` – updating the your server.
- `sudo apt install openssh-server -y` –installing the SSH.
- `sudo systemctl enable ssh` – enabling the SSH to your unit. (SERVER/DESKTOP)
- `sudo systemctl start ssh` – starting the SSH to your unit. (SERVER/DESKTOP)

Bridged makes the server behave like a separate PC in your LAN.

1. Find the Server's IP

Inside the VM, run:

```
Sample@UBUNTU-DESKTOP:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:9b:8c:ca brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.4/24 brd 192.168.1.255 scope global dynamic noprefixroute enp0s3
        valid_lft 83602sec preferred_lft 83602sec
    inet6 2001:4451:b00:2000:d76c:1af2:a10c:f36d/64 scope global temporary dynamic
        valid_lft 179068sec preferred_lft 83386sec
    inet6 2001:4451:b00:2000:a00:27ff:fe9b:8cca/64 scope global dynamic mngtmpadr proto kernel_r
        valid_lft 179068sec preferred_lft 92668sec
    inet6 fe80::a00:27ff:fe9b:8cca/64 scope link proto kernel_ll
        valid_lft forever preferred_lft forever
Sample@UBUNTU-DESKTOP:~$
```

From the desktop, connect to the server: `ssh username@192.168.1.120`

2. Connect from Desktop

From your host desktop:

```
Sample@UBUNTU-DESKTOP:~$ ssh Server@192.168.1.14
Server@192.168.1.14's password:
Welcome to Ubuntu 24.04.1 LTS (GNU/Linux 6.8.0-79-generic x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Mon Sep  8 11:24:38 PM UTC 2025

System load:          0.0
Usage of /:            5.1% of 48.91GB
Memory usage:         5%
Swap usage:           0%
Processes:            139
Users logged in:      1
IPv4 address for enp0s3: 192.168.1.14
IPv6 address for enp0s3: 2001:4451:b00:2000:a00:27ff:fe57:bc4a

Expanded Security Maintenance for Applications is not enabled.

136 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

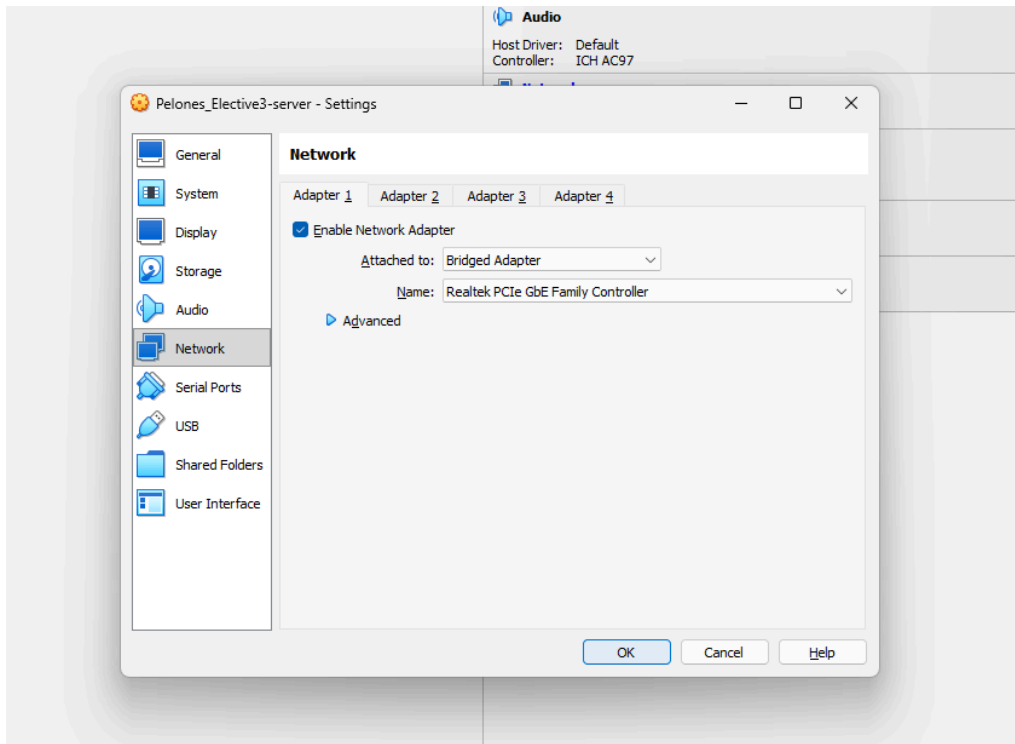
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon Sep  8 22:52:07 2025 from 192.168.1.4
Server@UBUNTU-SERVER:~$
```

Submission Requirements

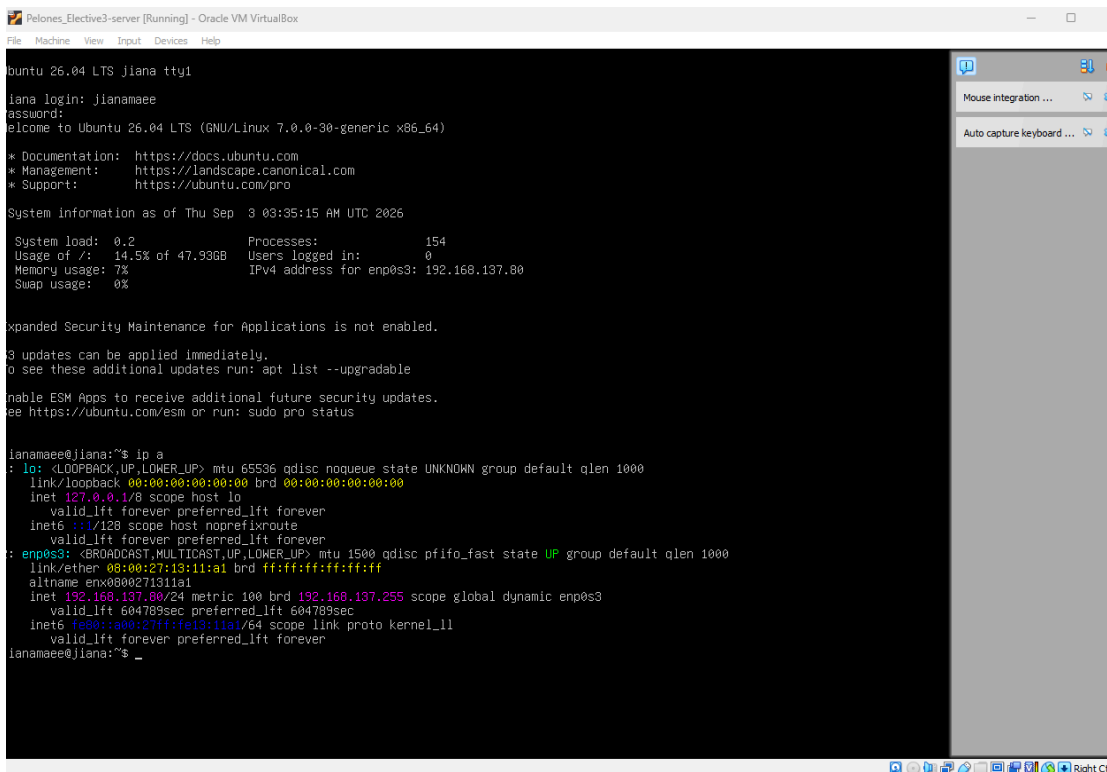
1. Screenshots (labeled clearly):

- VM network settings showing Bridged Adapter



-Open VM network through setting and attached it to bridged adapter.

- Server IP address (output of ip a or hostname -I)



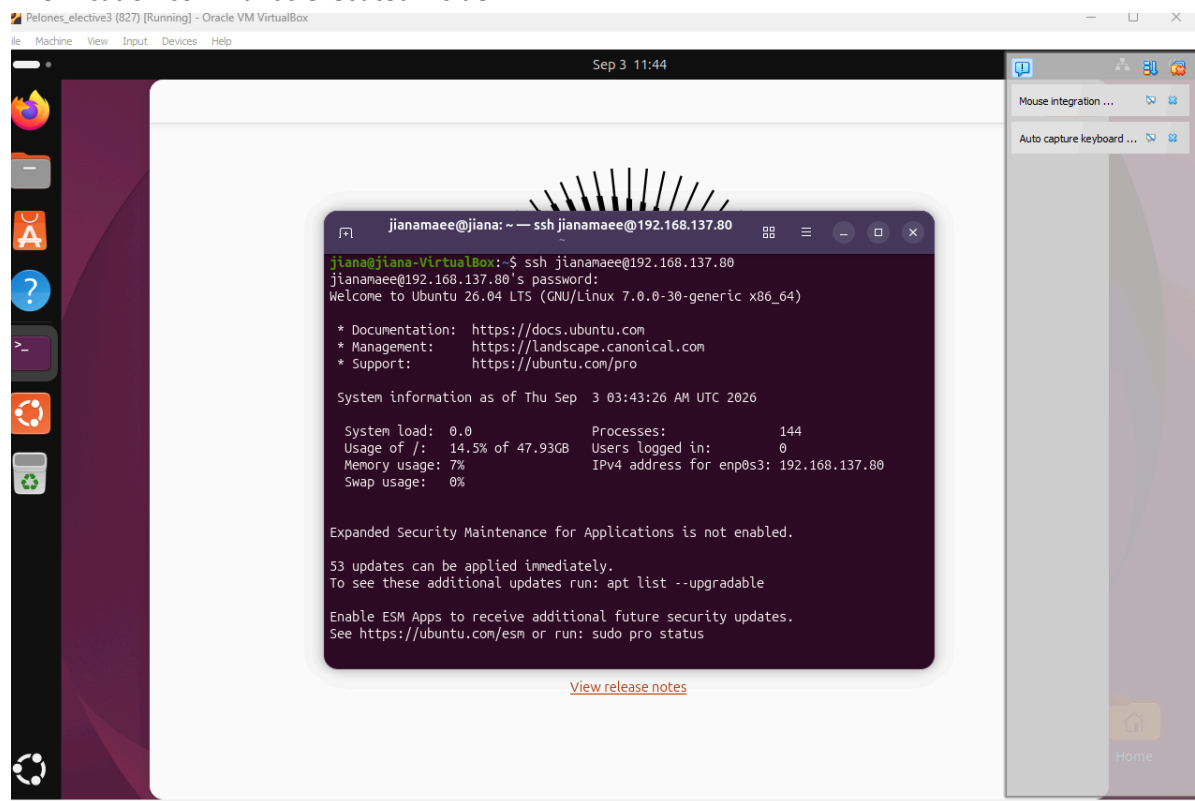
- SSH service status (sudo systemctl status ssh)

```
jianamaee@jiana:~$ sudo systemctl status ssh
[sudo: authenticate] Password:
• ssh.service - OpenBSD Secure Shell server
   Loaded: loaded (/usr/lib/systemd/system/ssh.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-03 04:03:51 UTC; 2min 55s ago
   Invocation: 2f828e5e8f244bb496f08fef4a6906bc
   TriggeredBy: • ssh.socket
   Docs: man:sshd(8)
        man:sshd_config(5)
   Process: 1289 ExecStartPre=/usr/sbin/sshd -t (code=exited, status=0/SUCCESS)
   Main PID: 1297 (sshd)
   Tasks: 1 (limit: 1567)
   Memory: 1.6M (peak: 2.4M)
   CPU: 27ms
   CGroup: /system.slice/ssh.service
           └─1297 "sshd: /usr/sbin/sshd -D [listener] 0 of 10-100 startups"

Sep 03 04:03:51 jiana systemd[1]: Starting ssh.service - OpenBSD Secure Shell server...
Sep 03 04:03:51 jiana systemd[1]: Started ssh.service - OpenBSD Secure Shell server.
Sep 03 04:03:51 jiana sshd[1297]: Server listening on 0.0.0.0 port 22.
Sep 03 04:03:51 jiana sshd[1297]: Server listening on :: port 22.
```

- Successful SSH connection from host

- Verification commands executed inside VM



2. Submission Format:

- Single PDF file
- Include name, student ID, and date
- All screenshots with brief explanation
- Upload to your portfolio website.

UBUNTU SERVER SSH REMOTE ACCESS LAB

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Course/Year/Section: BSCS3

Date Submitted: Sep 3, 2026

Instructor: Johneros P. Puyo

1 OBJECTIVES (5 POINTS)

Check all that apply:

- ✓ Create and configure a Virtual Machine with Ubuntu Server
- ✓ Configure Bridged Network Adapter for network access
- ✓ Determine and verify the server's IP address
- ✓ Install OpenSSH Server and enable the service
- ✓ Establish successful SSH connection from host machine
- ✓ Execute verification commands to confirm remote access

2 HARDWARE & SOFTWARE REQUIREMENTS (5 POINTS)

Item	Specification
Host Operating System	Windows 11
Virtualization Software (VirtualBox/VMware)	VirtualBox
Ubuntu Server Version	ubuntu-26.04-live-server-amd64
VM Name	Pelones_elective3
RAM Allocated	4GB
Storage Allocated	100GB
Network Mode	NAT
SSH Client Used	VirtualBox

3 VM CONFIGURATION

Setting	Value
VM Name	Pelones_Elective3-server
OS Type	Ubuntu (64-bit)
Memory	4000 MB

Hard Disk	100 GB
Network Adapter	Bridged

4. NETWORK CONFIGURATION

Commands Executed:

```

sudo systemctl status ssh
sudo apt update
sudo apt install openssh-server -y
sudo systemctl enable ssh
sudo systemctl start ssh
ip a
ssh jianamaee@192.168.137.80

```

Server IP Address: 192.168.137.80

Network Interface: enp0s3

6. REMOTE CONNECTION

Connection Details:

Parameter	Value
Server IP	192.168.137.80
Username	jianamaee
SSH Port	24